

1) Find out the application of Recursion

def :- Recursion is a technique where a function calls itself to solve a problem by breaking it into smaller subproblems

Applications

- Finding factorial of a number (Mathematical computation)
- Generating the fibonacci series
- Binary Search algorithm [Datastructure]
- Tree traversal (inorder, preorder, postorder)
- Tower of Hanoi problem
- Directory / File Searching in folders
- File System operations
- Dynamic programming
- Generating permutations / Combinations / subsets

2) Is recursion still used in the present programming era? So, where?

yes, recursion is still widely used in modern programming.

where it is used

- ↳ Data Structures (Trees and Graphs)
- ↳ Searching and Sorting algorithm (Quick Sort, Merge Sort)
- ↳ Backtracking problems (N-Queens, Sudoku, Solver)
- ↳ Dynamic programming
- ↳ File System navigation
- ↳ Artificial Intelligence (AI) and Compiler design
- ↳ Algorithms Libraries
- ↳ Modern frame works - React renders nested

Components recursively

3) what are the language that support recursive calls?

Many programming languages support recursion such as :

- C
- C++
- Java
- python
- Javascript
- C#
- Swift
- Kotlin
- Go
- Rust

Essentially, any language that allows a function to call itself (direct or indirectly) through another function) supports recursion